- Sequencing bias & errors
- Speed & RAM
- Data storage & visualization

Specific challenges:
(1) de novo sequencing

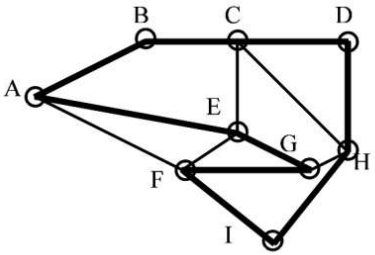


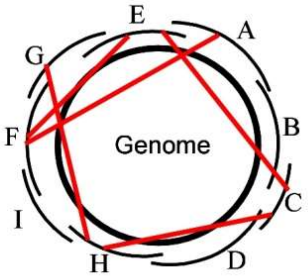
De novo assembly

Overlap Graph

Each read forms a *Node*

Edge exists between two nodes if the reads





Red/Thin: False Overlaps

Algorithm:

Step 1: Removing redundant edges, classify edges as required/optional

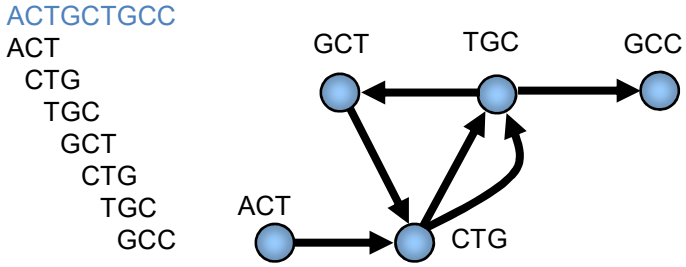
Step 2: Find the shortest walk which includes all required edges

Examples: Newbler, Celera, Arachne

De Bruijn Graph

Each read forms a *Node*

Edge exists between two nodes if the reads overlap



Algorithm:

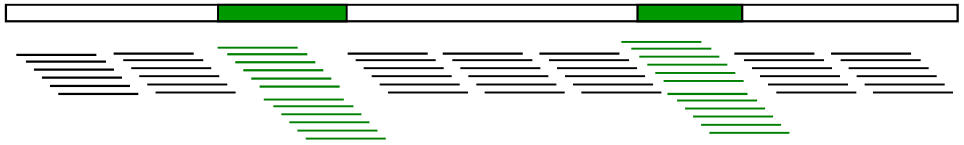
Step 1: Construct kmer hash table

Step 2: Build de Bruijn graph

Step 3: Simplify the graph and search Eulerian path

Examples: Euler, Velvet, Allpaths, Abyss, SOAPdenovo

Caveats



“Indeed, writing in Nature Methods earlier this year, Evan Eichler — a professor of genomics at the University of Washington — *criticized* the recent de novo assemblies produced by the Beijing Genome Institute (BGI), arguably one of the top three genomic institutes in the world.”

… compared two recent de novo human genome assemblies produced by the BGI and found that both were 16.2% shorter than the reference genome. In addition, the authors reported that 420.2 megabases of common repeats and 99.1% of validated duplicated sequences were missing from the assembled genome.

(Biotechniques.com, 04/28/2011)